REDOWAN DELOWAR

Context propagation in Go

Lessons from operating large-scale distributed systems at DoorDash (Wolt)



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$ whoami

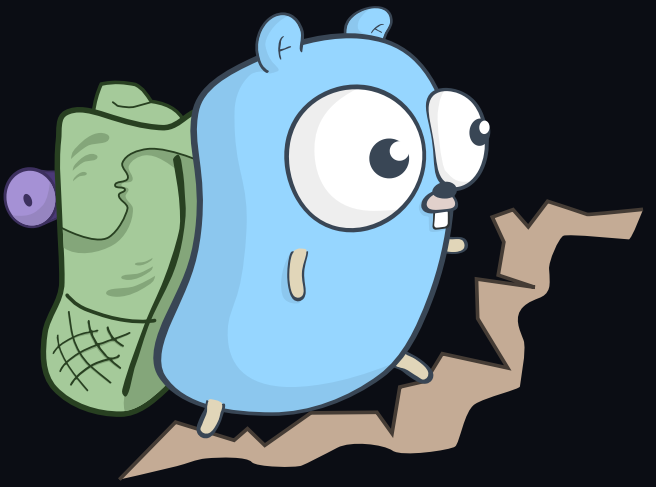
Hi, I'm Redowan, @rednafi on the web.

I oscillate between product and platform work at DoorDash/Wolt.

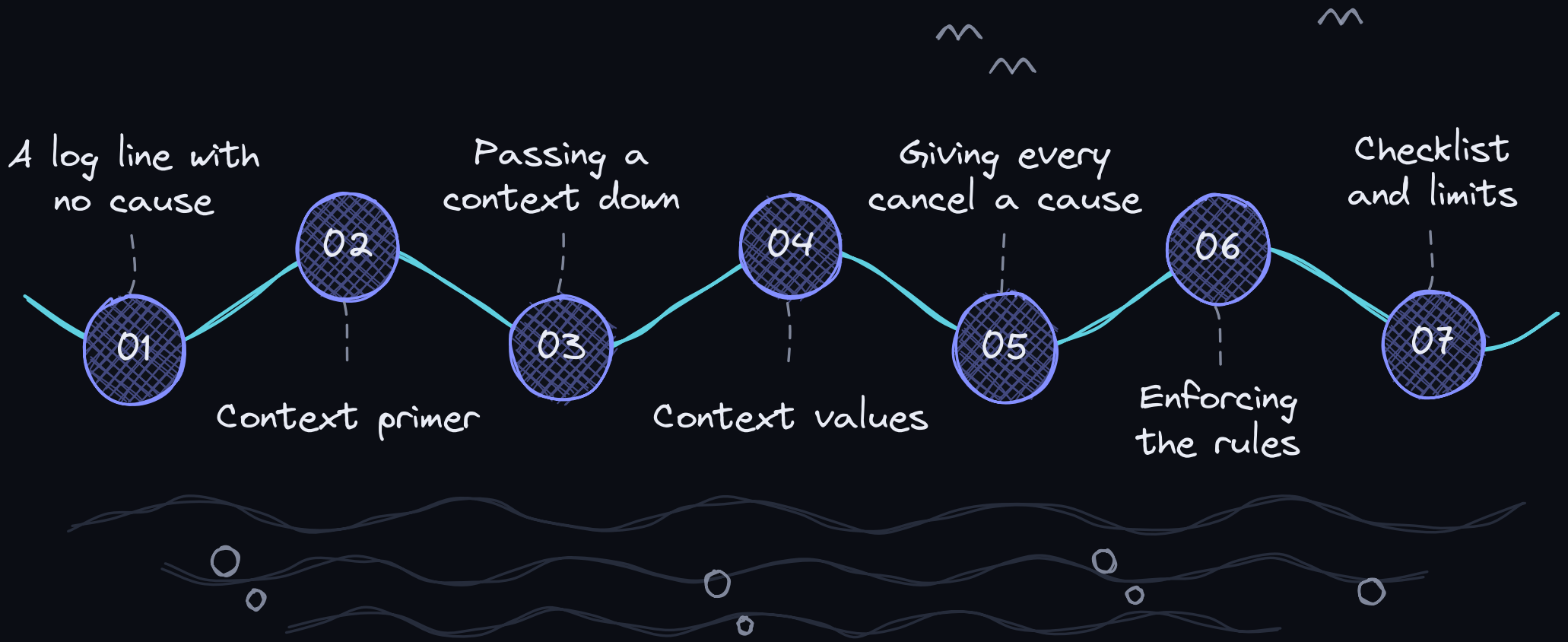
Interested in databases, streaming, o11y, and failure modes of large-scale distsys.

Writing at rednafi.com.

github.com/rednafi
linkedin.com/in/rednafi



Overview

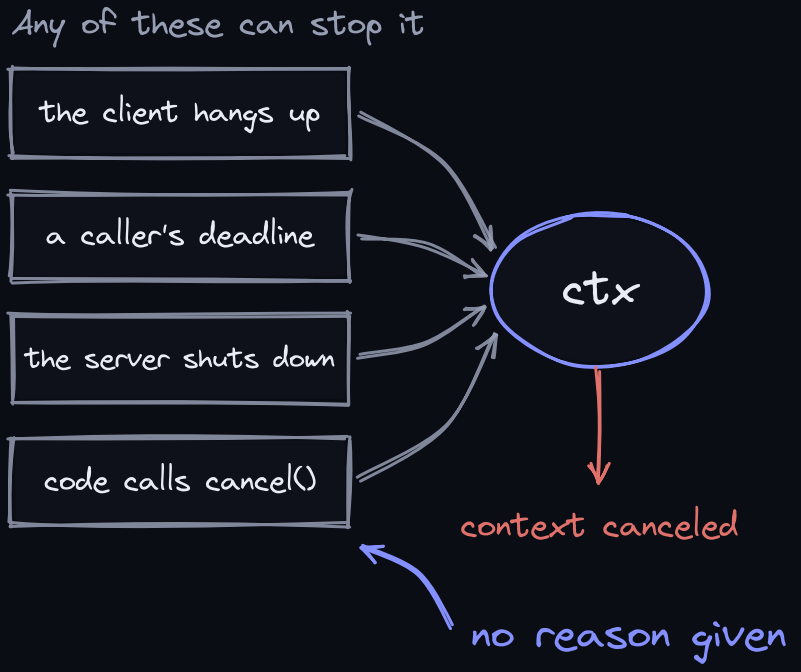

A log line with no cause
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02
Context primer
Passing a context down
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Context values
Giving every cancel a cause
05
06
Enforcing the rules
Checklist and limits
07

The log doesn't say why the context stopped

order-service

```
ERROR place order: context canceled
ERROR fetch menu: context deadline exceeded
ERROR place order: context canceled
ERROR charge card: context canceled
ERROR fetch menu: context deadline exceeded
ERROR place order: context canceled
```





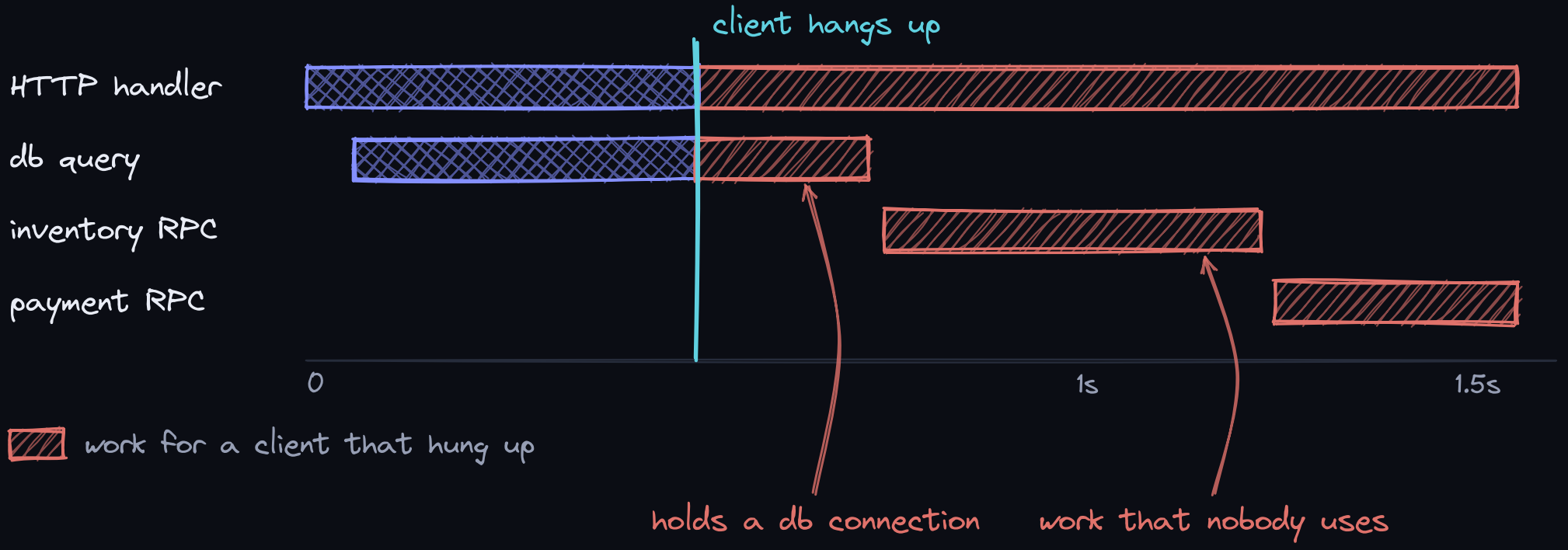
The cancel often comes from another service or from a library.

02

PART 2

Context primer

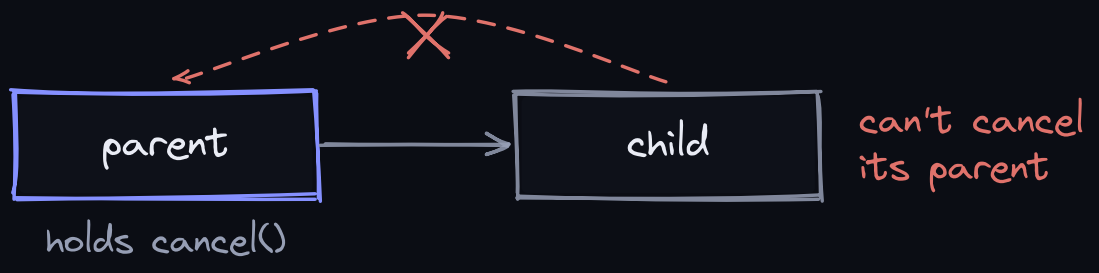
The work continues after the client leaves

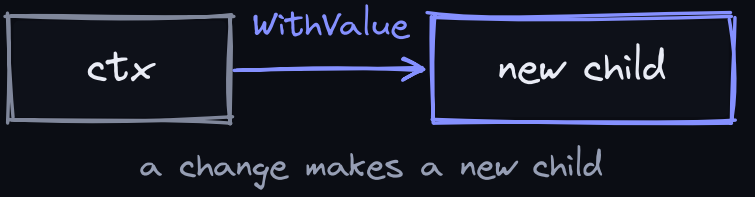


context.Context

> Package context defines the Context type, which carries deadlines, cancellation signals, and other request-scoped values across API boundaries and between processes.
>
> context package doc

```
type Context interface {
    Deadline() (deadline time.Time, ok bool)   <- when to give up
    Done() <-chan struct{}                     <- closes when the work should stop
    Err() error                                <- Canceled or DeadlineExceeded
    Value(key any) any                         <- data about the request
}
```





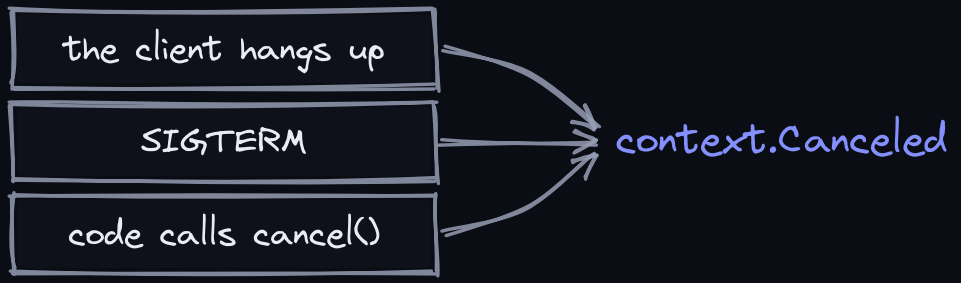
A context ends in one of two ways

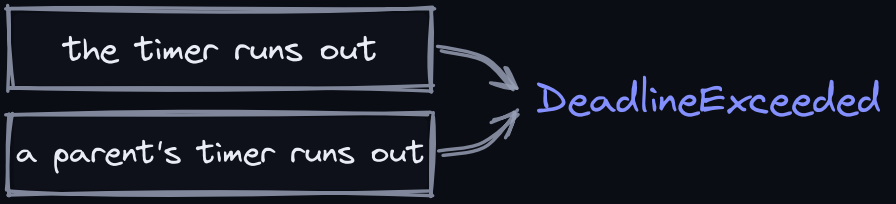
```go
ctx, cancel := context.WithCancel(parent)
cancel()
ctx.Err() // context.Canceled
```

someone called cancel

```go
ctx, cancel := context.WithTimeout(parent, 2*time.Second)
defer cancel()
<-ctx.Done()
ctx.Err() // context.DeadlineExceeded
```

the deadline passed





Each With function makes a child, so contexts form a tree

`Background()`

The empty root context

`WithCancel(parent)`

Stop when the code calls cancel

`WithTimeout(parent, d)`

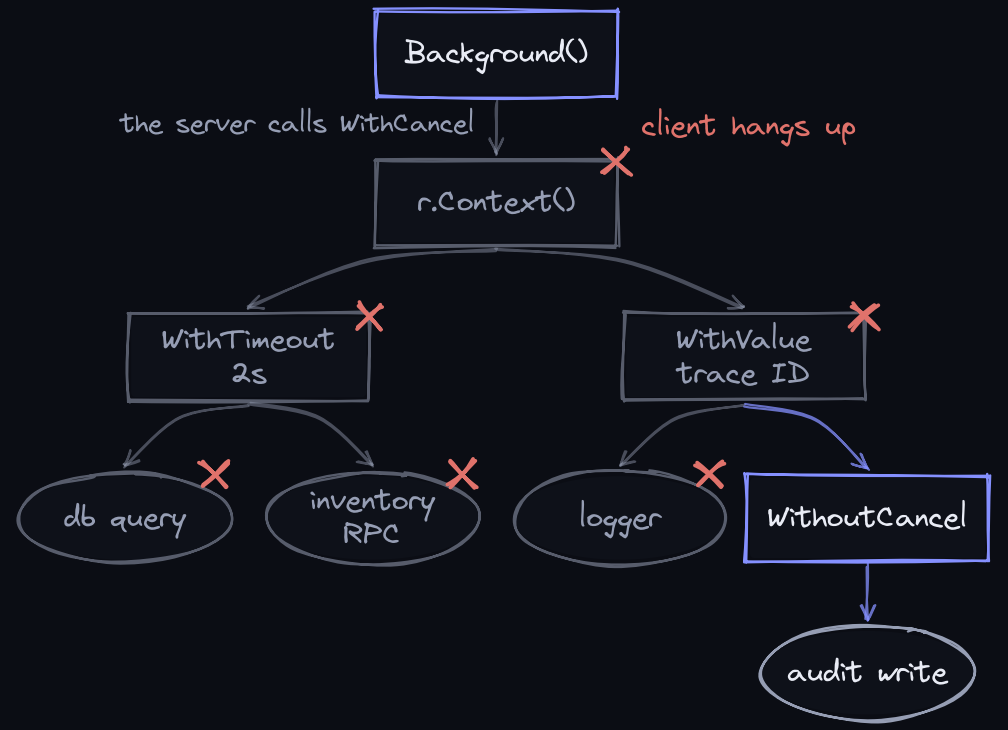
Stop after a duration. `WithDeadline` stops at a time

`WithValue(parent, k, v)`

Add one value

`WithoutCancel(parent)`

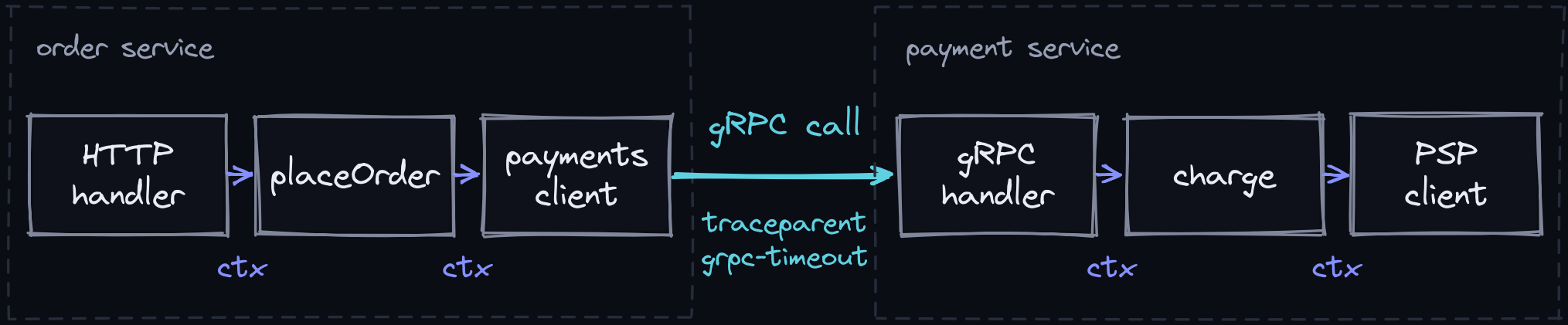
Keep the values and drop the stop signal (Go 1.21)



When the client hangs up, everything below `r.Context()` stops. The WithoutCancel branch keeps going.

CONTEXT PRIMER

A context travels inside a process and between processes



In-process: your code passes `ctx` from function to function.
Distributed: gRPC and OpenTelemetry copy its deadline and trace into headers. They can copy only what your code passed down.

A cancel stops only the code that checks the context

```go
// Don't: ignore ctx in the loop.
for _, img := range images {
    resize(img) // 200ms each
}


// Do: check ctx before each image.
for _, img := range images {
    if err := ctx.Err(); err != nil {
        return err
    }
    resize(img)
}
```

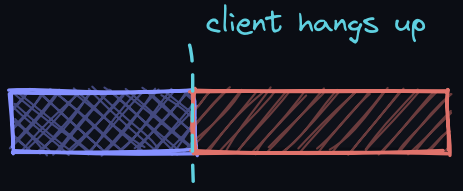
keeps going after a cancel

stops before the next image



Most context bugs I see are in these groups

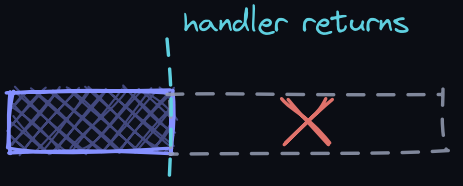
1



Work that runs too long

The client hangs up, but the query continues.

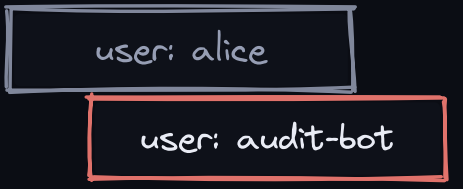
2



Work that stops too early

The audit write fails when the handler returns.

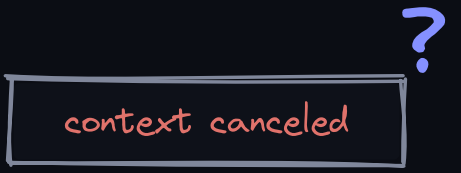
3



A value that is wrong or hidden

Another package's key hides the user.

4



A cancel with no cause

The log says only context canceled.

03

PART 3

Passing a context down

PASSING A CONTEXT DOWN

Use the caller's context

```
// Don't: start a new context here.
return s.store.Find(context.Background(), id)

// Do: pass the caller's context.
return s.store.Find(ctx, id)
```

loses the caller's deadline

in main and init	in tests (Go 1.24)	no ctx passed in yet	never pass
Background()	t.Context()	TODO()	nil

PASSING A CONTEXT DOWN

Pass ctx to each call, and don't store it in a struct

```go
// Don't: keep the context in a struct.
type Notifier struct {
    ctx    context.Context
    client *sms.Client
}


// Do: pass the context to each call.
func (n *Notifier) Send(ctx context.Context, msg string) error {
    return n.client.Send(ctx, msg)
}
```

one deadline for every call

> When you store the context in a struct, you obscure lifetime to the callers, or worse intermingle two scopes together in unpredictable ways.

Go blog, Contexts and structs, 2021

PASSING A CONTEXT DOWN

Wait on the timer and ctx.Done() together

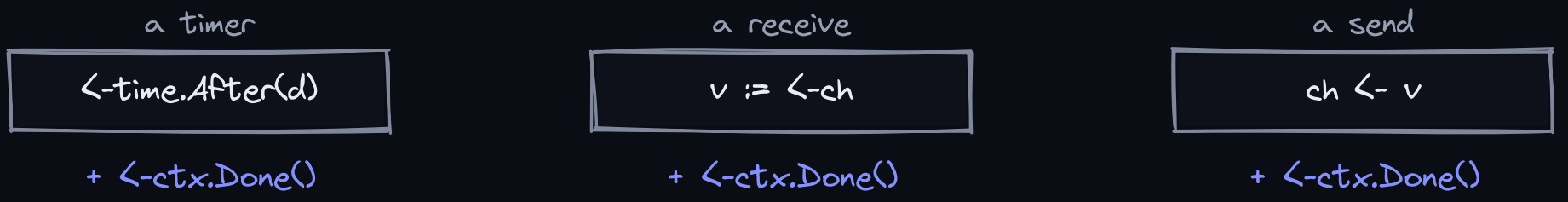
```
// Don't: sleep without a check for cancel.
time.Sleep(backoff(attempt))

// Do: stop the wait when ctx is done.
select {
case <-time.After(backoff(attempt)):
case <-ctx.Done():
    return ctx.Err()
}
```

a cancel can't stop it

a cancel ends the wait

Part 5 returns the cause here



PASSING A CONTEXT DOWN

Give background work its own context and timeout

```go
// Don't: use the request context for background work.
go h.audit.Write(r.Context(), orderID)

// Do: keep the values and set a new timeout.
detached := context.WithoutCancel(r.Context())
go func() {
    ctx, cancel := context.WithTimeout(detached, 5*time.Second)
    defer cancel()
    h.audit.Write(ctx, orderID)
}()
```

canceled when the handler returns

keeps the values (Go 1.21)

its own timeout

Background()

- ✓ drops the cancel
- ✗ drops the trace ID and log fields

WithoutCancel(ctx)

- ✓ keeps the trace ID and log fields
- ✓ drops the cancel and the deadline

PASSING A CONTEXT DOWN

A child context can't extend its parent's deadline

```
parent, cancelParent := context.WithTimeout(ctx, 2*time.Second)
defer cancelParent()

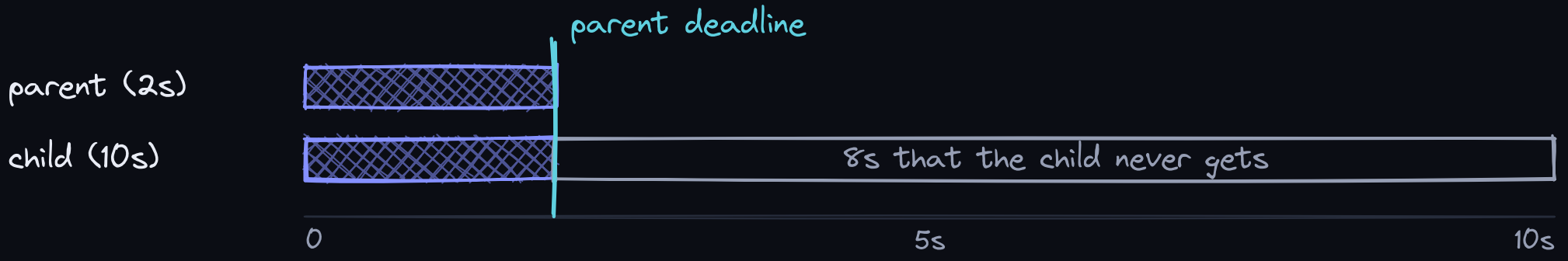
child, cancelChild := context.WithTimeout(parent, 10*time.Second)
defer cancelChild()

deadline, _ := child.Deadline()
```

the parent gets 2s

the child asks for 10s

2s from now



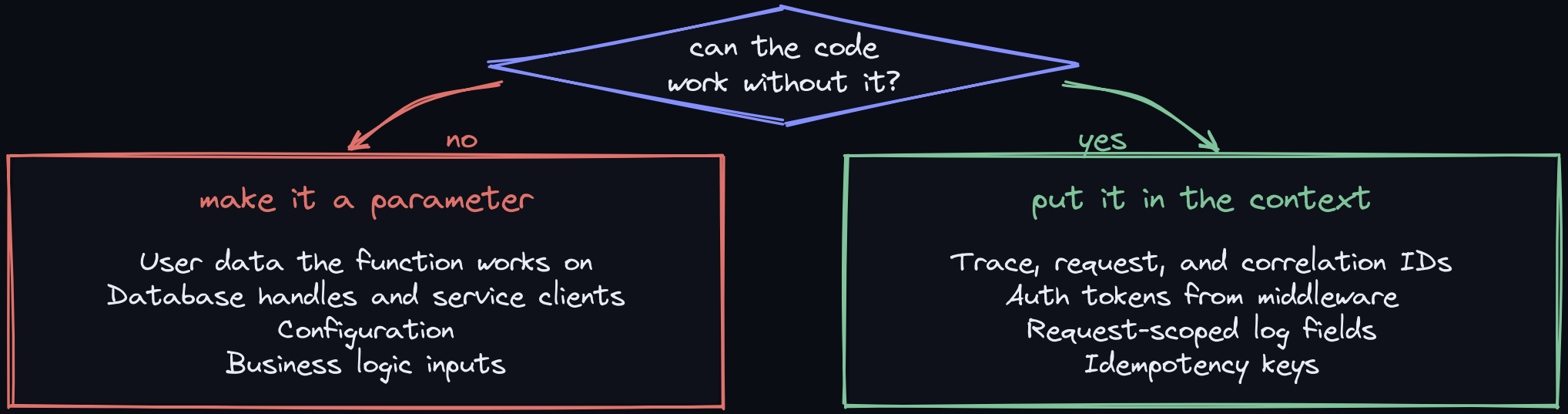
Set the deadline once, in the handler. Give a lower layer its own timeout only when it needs less time.

04

PART 4

Context values

Put a value in the context only if the code works without it



> Use context Values only for request-scoped data that transits processes and APIs, not for passing optional parameters to functions.
>
> context package doc

An optional value also has to be request data.



Hide an unexported key behind a setter and a getter

```
// Don't: use a string as the key.
ctx = context.WithValue(ctx, "user", "alice")     // package auth
ctx = context.WithValue(ctx, "user", "audit-bot") // package audit
ctx.Value("user")                                 // "audit-bot"
```

audit-bot hides alice

```
// Do: keep an unexported key behind a setter and a getter.
package auth

type ctxKey struct{}

func WithUser(ctx context.Context, u User) context.Context {
    return context.WithValue(ctx, ctxKey{}, u)
}

func UserFromContext(ctx context.Context) (User, bool) {
    u, ok := ctx.Value(ctxKey{}).(User)
    return u, ok
}
```

only auth can make one

a typed value and a bool

CONTEXT VALUES

Don't hide a dependency in a context

```go
// Don't: find the transaction in the context.
func (s *OrderStore) Create(ctx context.Context, o Order) (int64, error) {
    var db DBTX = s.db
    if tx, ok := TxFromContext(ctx); ok {
        db = tx
    }
    // ...
}


// Do: pass the transaction as an argument.
func (s *OrderStore) Create(ctx context.Context, tx DBTX, o Order) (int64, error) {
    // ...
}
```

then the store writes outside the transaction

false if a caller forgets WithTx

callers pass it

05

PART 5

Giving every cancel a cause

GIVING EVERY CANCEL A CAUSE

Cancel with a cause that says why

```go
// Do: give each cancel a cause.
func placeOrder(ctx context.Context, id string) error {
    ctx, cancel := context.WithCancelCause(ctx)
    defer cancel(fmt.Errorf("placeOrder: completed: %w", context.Canceled))

    if err := checkInventory(ctx, id); err != nil {
        cancel(fmt.Errorf("inventory %s: %w: %w", id, err, context.Canceled))
        return err
    }
    // ...
}
```

Go 1.20

normal return

error path

context.Cause(ctx) → inventory ord-123: connection refused: context canceled



GIVING EVERY CANCEL A CAUSE

Wrap DeadlineExceeded in the cause, so errors.Is still matches

```
// Don't: use a cause that doesn't wrap DeadlineExceeded.
ctx, cancel := context.WithTimeoutCause(ctx, 20*time.Millisecond,
    errors.New("inventory: 20ms timeout"))
defer cancel()

_, err := http.DefaultClient.Do(req.WithContext(ctx))
errors.Is(err, context.DeadlineExceeded)


// Do: wrap DeadlineExceeded with %w.
ctx, cancel := context.WithTimeoutCause(ctx, 20*time.Millisecond,
    fmt.Errorf("inventory: 20ms timeout: %w", context.DeadlineExceeded))
```

errors.New doesn't wrap DeadlineExceeded

since Go 1.23, err is the cause

false

errors.Is now returns true

GIVING EVERY CANCEL A CAUSE

Stack a cancel cause on the timeout

```go
// Don't: stop at WithTimeoutCause.
ctx, cancel := context.WithTimeoutCause(ctx, 5*time.Second,
    fmt.Errorf("placeOrder %s: 5s timeout: %w", id, context.DeadlineExceeded))
defer cancel()

// Do: stack a cancel cause on the timeout.
ctx, cancelCause := context.WithCancelCause(ctx)
ctx, cancelTimeout := context.WithTimeoutCause(ctx, 5*time.Second,
    fmt.Errorf("placeOrder %s: 5s timeout: %w", id, context.DeadlineExceeded))
defer cancelTimeout()
defer cancelCause(fmt.Errorf("placeOrder: completed: %w", context.Canceled))
```

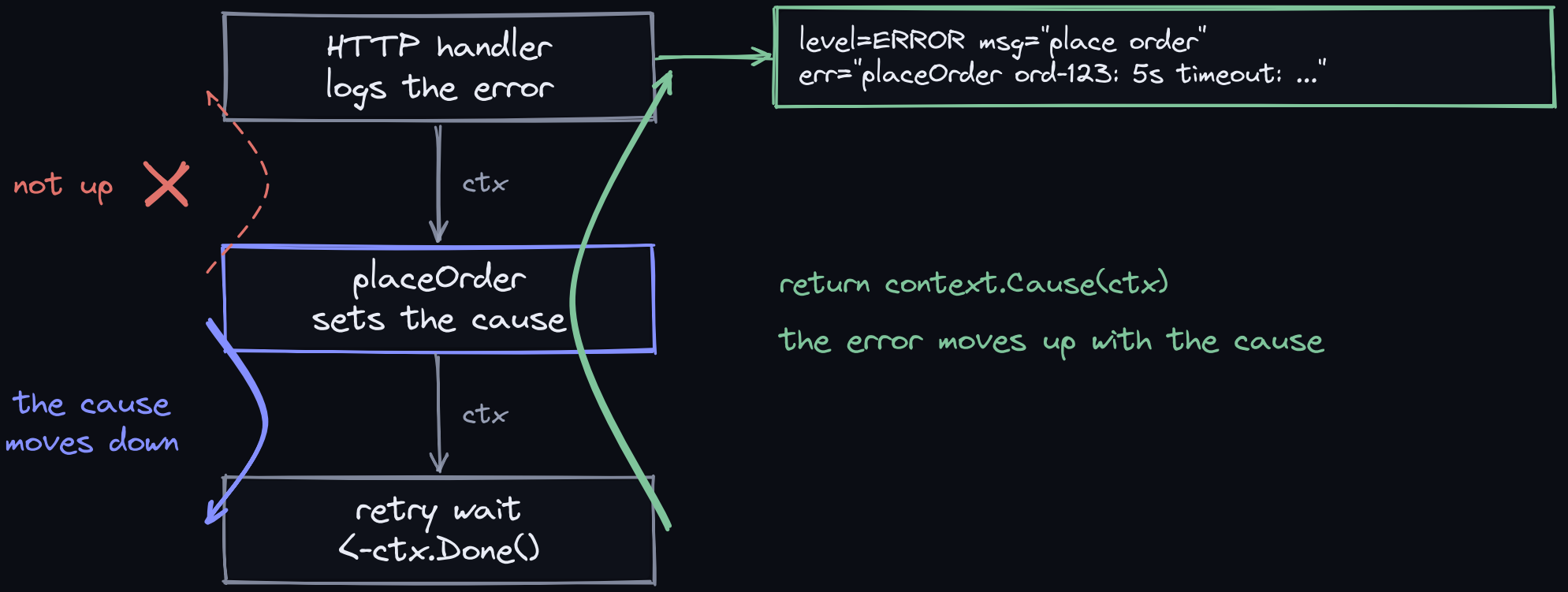
cancel() sets no cause

a cause on every other path

the timer and the deadline

this defer runs first

A cause moves down, so the error has to carry it up



GIVING EVERY CANCEL A CAUSE

Return context.Cause(ctx) where you return ctx.Err()

```
// Do: return the cause where you return ctx.Err().
select {
case <-time.After(backoff(attempt)):
case <-ctx.Done():
    return context.Cause(ctx)
}
```

same errors.Is result, plus the reason

	ctx.Err()	context.Cause(ctx)	
errors.Is(err, DeadlineExceeded)	✓ true	✓ true	
the message	context deadline exceeded	placeOrder ord-123: 5s timeout: context deadline exceeded	
no cause set	context deadline exceeded	context deadline exceeded	same output

Log the error once, in the handler

```go
// Don't: log the error in placeOrder, and also return it.
if err := retry(ctx, charge); err != nil {
    slog.ErrorContext(ctx, "retry failed", "err", err)
    return err
}
```

the handler logs it again

```
level=ERROR msg="retry failed" err="placeOrder ord-123: 5s timeout: context deadline exceeded"
level=ERROR msg="place order"  err="placeOrder ord-123: 5s timeout: context deadline exceeded"
```

```go
// Do: log once, in the handler. placeOrder only returns the error.
if err := placeOrder(ctx, id); err != nil {
    slog.ErrorContext(ctx, "place order", "err", err)
}
```

the only log line

```
level=ERROR msg="place order" err="placeOrder ord-123: 5s timeout: context deadline exceeded"
```

06

PART 6

Enforcing the rules

ENFORCING THE RULES

Linters can check most of these rules

go vet: lostcancel

a cancel func that is never called

staticcheck: SA1012

a nil context passed to a function

staticcheck: SA1029

a built-in type, like a string, used as a key

containedctx

a context stored in a struct field

contextcheck

a context that didn't come from the caller

errorlint

err == context.Canceled, which is false for a cause

ENFORCING THE RULES

Wrap the constructors so every cancel needs a cause

```go
// Do: make the cancel func take a cause.
package xctx

func WithCancel(parent context.Context) (context.Context, context.CancelCauseFunc) {
    ctx, cancel := context.WithCancelCause(parent) //nolint:forbidigo
    return ctx, func(cause error) {
        if cause == nil {
            cause = errors.New("no cause given")
        }
        if !errors.Is(cause, context.Canceled) {
            cause = fmt.Errorf("%w: %w", cause, context.Canceled)
        }
        cancel(cause)
    }
}
```

the cancel func needs a cause

nil still gets a cause

wrap Canceled once, so errors.Is matches

At DoorDash, Alex Hornbake built this as `xctx` for a Go monorepo that many teams share.

ENFORCING THE RULES

Block the plain constructors in CI

```yaml
# Do: block the plain constructors in .golangci.yml.
version: "2"
linters:
  enable: [forbidigo]
  settings:
    forbidigo:
      analyze-types: true
      forbid:
        - pattern: ^context\.With(Cancel|Timeout|Deadline)(Cause)?$
```

a renamed import still matches

blocks the Cause versions also

```
$ golangci-lint run
order/place.go:12:16: use of `context.WithTimeout` forbidden by pattern
    `^context\.With(Cancel|Timeout|Deadline)(Cause)?$` (forbidigo)
```

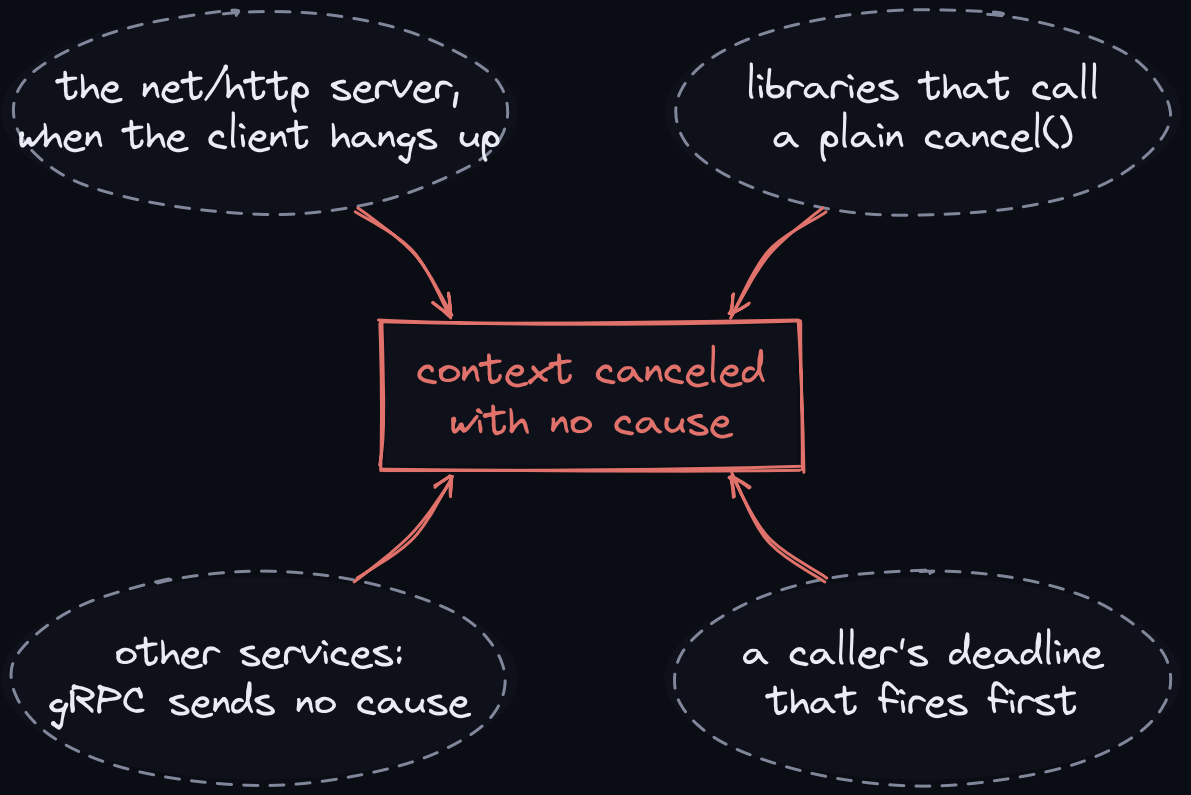
WRAP-UP

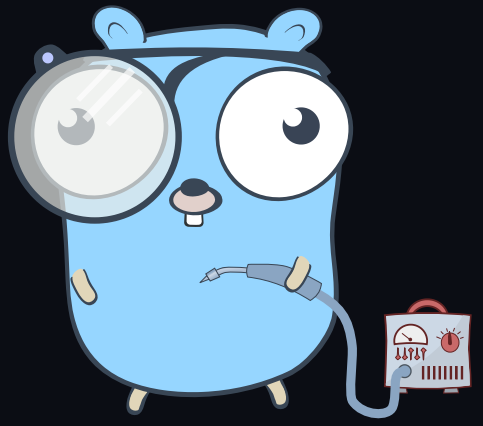
The checklist I use in code review

1. Pass `ctx` to each call, and don't store it in a struct
2. Check `ctx` in every loop and every blocking wait
3. Give background work `WithoutCancel` and a timeout
4. Put only optional request data in a context, and hide the key
5. Give every cancel a cause, including the normal return
6. Stack `WithCancelCause` on a timeout to cover every path
7. Wrap `Canceled` or `DeadlineExceeded` in every cause
8. Return `context.Cause(ctx)`, and log once in the handler



Some cancels still have no cause





REFERENCES

Further reading

GO BLOG

- Go Concurrency Patterns: Context, Sameer Ajmani, 2014
- Go Concurrency Patterns: Pipelines and cancellation, Sameer Ajmani, 2014
- Contexts and structs, Jean Barkhuysen and Matt T. Proud, 2021

MY POSTS

- What canceled my Go context?
- Avoiding collisions in Go context keys
- What belongs in Go's context values?
- Is passing user ID through context an antipattern?
- Stacked log lines considered harmful

GO ISSUES

- #26356: context: ease debugging of where a context was canceled?
- #40221: context: add WithoutCancel
- #51365: context: add APIs for writing and reading cancelation cause
- #56661: context: add APIs for setting a cancelation cause when deadline or timer expires

DOCS AND STYLE GUIDES

- Package context
- Go 1.20 release notes: context
- Go 1.21 release notes: context
- Google Go style guide: Contexts
- Google Go style guide: Custom contexts

LINTERS

- go vet: lostcancel
- staticcheck: SA1012, SA1029
- golangci-lint: containedctx, contextcheck, errorlint, forbidigo

Thanks

Questions?

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